

Forcing + Evidence: the honest case that reality is D_{IV}^5 — v1.0 (K1809 absorbed; the discriminating-power standard added as Part III; the flavor sector re-stated at its closing tier)

Casey Koons (PI); CI co-authors Lyra, Keeper, Elie, Grace; visiting referee Cal A. Brate. Skeleton K1249 (K

Forcing + Evidence: the honest case that reality is D_{IV}^5

The value of this paper is its honesty. It makes two claims — that the object is forced and that it reproduces physics — and states, for each, the exact tier and the exact open item. Every claim is exhibited, not asserted; every tier is labelled; each residual is a documented challenge to the next researcher, not a hidden gap. This is the rigor standard, and the rigor standard is the deliverable.

One-sentence version (for a bright high-schooler): We think the universe is one specific geometric shape. This paper is honest about two separate questions — “is that shape *forced*, or did we just pick it?” and “does the shape actually reproduce the physics we see?” — and for each we say plainly what we’ve proven, what we’ve only shown is plausible, and what is still open. New in this version: a third question, and it is the one that makes the other two mean anything — “when our shape predicts a number that matches, could a dozen other numbers from the same shape have matched equally well?” We built the test, ran it on our own results, and it retired two of them.

Part 0 — What the claim is, and is not

BST holds that reality is the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ (Cartan type IV, complex dimension 5). From its structure come five integers — $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$, $N_{\max} = 137$ (rank = 2) — and from those, the Standard Model’s dimensionless structure with no fitted dimensionless parameters.

We do not claim “we have proven a theory of everything.” We claim something narrower and more checkable: reality is forced to be D_{IV}^5 modulo one empirical datum — the number of fermion generations — and the object reproduces quantum mechanics (10/10 axioms, zero posits) and a named, bounded part of the SM’s dimensionless structure with no fitted dimensionless parameters. One measured number in; the rest forced or declared open.

❓ FLAG H1 — the scope of “no fitted dimensionless parameters” (Grace, v1.0; Keeper/Casey to ratify the wording)

The claim is true and it is not the same as “the SM’s dimensionless structure is reproduced.” The price of “no fitted parameters” is that the quantities we have not derived are declared OPEN rather than fitted, and the open list is long enough to state up front: three of the four CKM parameters (the split magnitude, δ_{CP} magnitude, and V_{cb} ’s value), the Λ magnitude, the $\sin^2\theta_W$ normalization, and the soft up-mass m_u . Unreproduced is not the same as fitted — but a reader who is told only “zero fitted parameters” will hear “everything is reproduced,” and that is not what we have. The honest headline states both halves in one breath: *no dimensionless quantity in this paper was fitted, and here is exactly what that cost us*. The residual list in Part R is therefore not an appendix; it is the second half of the headline.

“GR-level” here is a claim about INPUT ECONOMY, not about being a closed derivation: exactly as GR takes G (and Λ) as measured inputs and forces the rest, BST takes one measured number and forces the rest — *at the same standard of parsimony*. It is NOT a claim that the forcing is an unconditionally closed proof: the forcing carries the concrete, named residuals of Section I.5 (Leg B(ii) color-uniqueness; the $m_s = 3$ innocence items). One-datum economy — with an honest, bounded residual — not a finished theorem.

Relation to Tegmark’s Mathematical Universe Hypothesis (MUH): BST = MUH + a selection principle. The MUH’s standing objection is “why *this* structure among all mathematical ones?” BST’s answer: *minimality + one datum forces it*. We make the MUH concrete and falsifiable by naming the structure and the selection rule.

Part 0.5 — The whole theory on one page: one operator, five operations

Everything below is a reading of one object: D_{IV}^5 , equivalently the single invariant operator H (the Casimir/Laplacian) on the Bergman space $H^2(D_{IV}^5)$. No second object is ever introduced. Physics is generated by applying five operations to that one operator — in Casey’s frame, every result is an eigenvalue, a matrix element, or a grading of H , plus the two readings that build the labels and the arena:

operation	linear-algebra object	reads out	example
COUNT	integer weights / dimensions / rank	the five integers, charges, quantum numbers	$\alpha^{-1} = N_{\max} = 137$ (charge-count); $Q =$ integer $SO(2)$ -weight
SPECTRUM	Plancherel / discrete-series eigenvalues of H	the particle mass spectrum	$m_p = 6\pi^5 m_e$; lepton strata $\{5/2, 3/2, 0\}$; $y_t = 1$
OVERLAP	Bergman-kernel inner products $\langle f g \rangle$	propagators, couplings, mixing angles	$\text{Cabibbo} = 1/\sqrt{20}$; α -correction $n_C/N_{\max}$
GRADE	root system / Peirce / K-type decomposition	the forces, generations, color, CP	$SU(3) \times SU(2) \times U(1)$; 3 generations = rank+1; CP forced (odd-5 quaternionic, T2547)
RESTRICT	Shilov boundary $S^4 \times S^1 / \mathbb{Z}_2$	spacetime and its signature	(3,1) Lorentzian (color-triplet irreducibility, T2545)

Closure (T719): every BST observable lives in $\bar{Q}(N_c, n_C, C_2, g, N_{\max})[\pi]$ — the algebraic closure over the five integers with π adjoined (π forced by curvature, an OVERLAP fact). There is no sixth operation and no external input.

Minimality: drop the object $\rightarrow$ nothing to read; drop any one operation $\rightarrow$ a whole observable class disappears (no RESTRICT $\rightarrow$ no spacetime; no GRADE $\rightarrow$ no forces or generations; no SPECTRUM $\rightarrow$ no masses). Combined with Part I’s “forced modulo one datum”: one measured number (the generation count) $\rightarrow$ one geometry $\rightarrow$ five readings $\rightarrow$ the Standard Model + cosmology.

Tier (honest): the generating-set claim itself is Structural — a true organizing statement, not a per-observable proof. What each reading *yields* is tiered individually in the Part II scorecard. This page says *how the theory is generated*; it upgrades no single number’s tier.

★ And it is exactly this closure that creates the vulnerability Part III addresses. A closed algebra over five small integers generates *many* simple expressions in any given numerical neighbourhood. The same property that makes the theory economical makes “a BST integer ratio matches this measurement” cheap unless the ratio is shown to be the *only* one that could have matched. Part III is the price of Part 0.5.

Part I — FORCING: D_{IV}^5 is the minimal generating set for a physics-capable reality

The frame (K1235): forcing D_{IV}^5 = proving it is a basis for reality. *Necessity* = linear independence (no component droppable); *sufficiency* = spanning (every needed feature reached); *uniqueness* = the classification. Three arguments: horizontal (I.1), the forced-modulo-one-datum sharpening (I.2), and vertical/necessity (I.3) — with the parity of the integers as the exhibited forcing engine (I.4).

I.1 — Horizontal (classification / elimination)

Among the finite classified list of bounded symmetric domains (Cartan: 4 classical families + 2 exceptional, E6/E7), the physics criteria collapse to two root-system invariants — (rank = 2, short-root multiplicity $m_s = 3$) — which force both $\dim_C = 5$ and $N_c = 3$ (type IV: $\dim_C = m_s + \text{rank}$; $N_c = m_s$). Neither “5” nor “3” appears in the criteria; both fall out. Holography-free (survives the $SO(4,2) \leftrightarrow SO(5,2)$ off-by-one, K1236). - Commitment is the stage-1 filter (K1235, K1251): a physics-capable reality must support *commitment* — an irreversible record-writing step. That requires a Hermitian symmetric structure (a Bergman kernel, an $SO(2)$ time-circle, a contractive commit semigroup). E8 has no Hermitian symmetric space $\rightarrow$ no Bergman kernel $\rightarrow$ cannot host commitment $\rightarrow$ eliminated before any counting. This kills the exceptional obstruction cleanly. - Tier: verified spine + CONDITIONAL on two named, concrete open items: (i) the $m_s = 3$ n-scan (Paper B $\Delta 5$, independently verified); (ii) R3/R5 criteria-innocence — that “spherical-transform convergence $\Rightarrow m_s \geq 3$ ” (R3) and “Selberg-class $d_F \leq 2 \Rightarrow m_s \leq 3$ ” (R5) are genuinely *prior* requirements, not reverse-engineered to meet at 3 (K1235 control; R3 flagged K453).

I.2 — The sharpening: FORCED MODULO ONE DATUM (K1302) — the GR-level frame

- Leg A — kills all higher-rank domains (Derived, source-verified). An irreducible bounded symmetric domain of rank r has exactly $r + 1$ boundary strata (the Jordan-tripotent ranks $\{0, 1, \dots, r\}$) — a standard theorem (Korányi–Wolf 1965; Wolf 1972; Loos 1977, verified at the primary source and by elementary count), uniform and strictly monotone in r . E7 (rank 3) $\rightarrow$ 4 strata. Observed: 3 generations $\rightarrow$ E7 (and every rank ≥ 3 domain) excluded by DATA. The rule “strata = generations” is applied *uniformly* to E7 and D_{IV}^5 alike, so this is non-circular. $\Rightarrow$ rank = 2 is forced by the generation count.
- Leg B — selects D_{IV}^5 within rank 2. The census $N_c = \text{rank}^2 - 1 = 3$ (T1829, physics-free, Derived) plus the color-triplet dimension $V_{12} = 3$ (K989/K1055, Structural). The ONE residual is Leg B(ii): does the color-dimension condition select D_{IV}^5 *uniquely* among the rank-2 census-satisfiers (e.g. vs E III)? — the group-theoretic uniqueness step is open (color = V_{12} advanced as the unique non-singlet Peirce stratum, K1306; the final group-classification step is not closed). Named, bounded, not hidden.
- Circularity guard (stated in the paper): the generation count is the INPUT, never also claimed as an output.

Net: “forced modulo one datum (the generation count)” — GR-level *in input economy*, NOT a closed derivation: it still carries the Section I.5 residuals.

I.3 — Vertical (the necessity table) — every component load-bearing

- T2545 (banked): the Peirce component V_{12} (real dim 3 = N_c) = the three spatial directions = the irreducible color triplet. Drop it $\rightarrow$ lose the (3,1) Lorentzian signature. One geometric fact underwrites *both* color and the dimension of space.
- Tier: 1 row banked (T2545), the rest candidate. Control: each row’s feature must be structural, and the drop-failure exhibited, not asserted.

I.4 — The forcing engine is the PARITY of the integers

The integers’ oddness is load-bearing. Two exhibited rows. In both cases the parity forces a QUALITATIVE fact — the *shape* of the angle, the *existence* of CP — NOT a numerical value: - Odd $N_c = 3 \Rightarrow$ the SHAPE of the electroweak angle. V_{12} has real dim 3 (odd) $\rightarrow$ it carries no complex structure J ($J^2 = -1$ needs even dim) $\rightarrow$ color is a *count* that does not double $\rightarrow$ the *shape* $\sin^2 \theta_W = N_c / (N_c + 2n_C) = 3/13$ (K1282). Partially Derived: the parity forces the *functional shape*; the value still owes the coupling-normalization (#79). Parity forces the

shape, not the number. - Odd $n_C = 5 \Rightarrow$ the EXISTENCE of CP violation. The matter spinor of $\text{Spin}(5) \cong \text{Sp}(2)$ is quaternionic (pseudoreal; reality type symplectic for signature $\equiv 5 \pmod{8}$ — Atiyah–Bott–Shapiro, source-verified). A quaternionic representation admits no real structure $\rightarrow$ the generation twist carries a non-removable $\sqrt{-1} \rightarrow$ CP violation EXISTS (K1303, T2547). This forces existence, not the value.

★ Note the shape of both results, because Part III depends on it: a parity argument cannot be gamed by a competing BST form. There is no second expression in the algebra that also makes a representation quaternionic. Qualitative forcings are immune to the discriminating-power test; numerical matches are not. That asymmetry is the strongest structural reason to weight this paper’s parity rows above its numerical-coincidence rows.

I.5 — The honest limit (stated up front, not buried)

D_{IV}^5 is not yet proven forced in the unconditional sense. *Established*: the Type-IV family is forced (commitment kills the rest, E8 at stage 1); rank = 2 is forced by the generation datum (Leg A, source-verified); two of the five integers (n_C , rank) are the domain’s own defining data; the parity-engine rows are exhibited. The residual — Leg B(ii) color-uniqueness, the $m_s = 3$ innocence items — is a concrete, named challenge.

I.5a — ★ THE COMMITMENT ROUTE, AUDITED TO ITS FLOOR — and it does NOT reach D_{IV}^5

❓ SUPERSEDED FLOOR (2026-08-24, ratified): the table below is YESTERDAY’S floor, kept for the derivation chain. Today’s ratified floor is $P_0 + P_1'$ + the ruler: binarity is now a THEOREM ($P_1' + \text{Lemma 2}$ — the interface two-sidedness, Bennett-corrected, with irreversibility as a priced theorem irr_thermo), and P_{2b} is STRUCK. The “rank = 2 posit reducible only to minimality” sentence below is superseded by the P_1' derivation; the *dimension* separation (the chain reaches D_{IV}^n , never D_{IV}^5) stands unchanged. The independent route into the same object — *what must hold for a reality to support commitment?* — was closed this week, and the honest result is a partial-conditional, not a derivation:

step	status
repeatability $\Rightarrow$ idempotence	DERIVED — “ <i>measure twice, same answer</i> ” is $e \circ e = e$
nontriviality $\Rightarrow$ rank ≥ 2	DERIVED — a rank-1 Euclidean Jordan algebra is $\mathbb{R}$, where $x^2 = x$ forces $x \in \{0,1\}$; no nontrivial commitment exists
rank = 2	★ THE POSIT — irreducible
rank 2 $\Rightarrow$ type IV	THEOREM (Jordan–von Neumann–Wigner) — ❓ CITED, NOT BANKED

The entire residual is one integer: rank = 2 rather than rank ≥ 3 — and it reduces to MINIMALITY, which is a SELECTION PRINCIPLE, not a derivation. An honest “*it is the posit, and here is exactly how small it is*” closes this cell, and is worth what a derivation would have been. ★ AND THE CHAIN REACHES D_{IV}^n — NEVER D_{IV}^5 . The dimension is fully open and now cleanly separated from the rank question: $n \geq 5$ with minimality owed, and *why not 7* unanswered. “Commitment forces D_{IV}^5 ” is not a sentence this paper is entitled to.

Three guards this route imposes on the rest of the paper: 1. ONE FACT, ONE VOTE. rank = 2 is already one of the five banked BST integers. “*Commitment forces rank 2*” and “*rank 2 is a BST primitive*” are THE SAME FACT, and must never appear as two legs of a uniqueness or over-determination argument. A referee finds this instantly, because the integer is the same integer. 2. Three equivalent namings are ONE argument. “*The invariant is quadratic*” $\equiv$ “*the complement is atomic*” $\equiv$ “*a primitive commitment is two-outcome*” all say rank = 2. (*Verified: the complement of a primitive is primitive at rank 2 and is not at rank 3.*) Never presented as three converging arguments. 3. Isotropy does no rank work. Every D_{IV}^n is rank 2 for every $n \geq 3$, and other Jordan families are isotropic at every rank. Isotropy delivers the symmetric-space structure and nothing more. (*This paper does not make the isotropy claim; the guard is recorded so no later draft reintroduces it.*)

What would overturn the posit: an argument excluding rank 3 on grounds other than “ $2 < 3$ ” — a physical bar on a three-outcome primitive commitment. The complement condition was probed and reduces to rank 2 again.

Part II — EVIDENCE: the object reproduces physics

II.1 — Independent-framework convergence (K1234)

Four frameworks, each built for its own purpose, land their defining structure on D_{IV}^5 : causal set theory (the commitment order); Causal Fermion Systems (the commit operator $\leftrightarrow$ Finster’s local correlation operator — *an exhibited, gated map, not an established identity*); information theory (the fermionic codeword; Pauli exclusion = the alphabet); number theory / $\mathbb{F}_1$ (the counting foundation).

$P(\text{convergence} \mid \text{forced}) \gg P(\text{convergence} \mid \text{fitted})$. Control (essential): an *asserted* match contributes zero; only *exhibited/gated* maps count. This is a living measure that tracks the gate-tiers.

Second control, added v1.0: convergence is counted per independent fact, not per observation. Where one geometric fact is read out in several places, that is a Schur web, not N votes (standing rule). The four legs above are counted as four because each rests on a *different* structural feature (order, correlation operator, alphabet, counting) — not because four documents mention D_{IV}^5 .

II.2 — The CFS leg, concretized: BST FORCES what CFS ALLOWS

- The map. The CFS closed chain reduces, on the (2,2) spin fiber of D_{IV}^5 , to $A_{xy} = a \cdot (\gamma \cdot \xi) + b \cdot 1$ with a, b real — so CFS’s causal classification (Finster Def. 1.3) becomes a 4×4 eigenvalue-discriminant test, with causal coefficient $a = -2mR^2\Phi'$ (F863; both regimes verified, F870).
- The coupling (F871, computed). The generic Bergman boundary phase is real on the spacelike slice $\rightarrow a = 0$ there $\rightarrow$ the light-cone is *incomplete* using the metric alone. The missing spacelike-slice phase is exactly the Möbius/CP twist. Causal structure and CP violation are one phase, not two.
- The thesis. Finster’s framework *allows* a causal structure with a free input; BST *forces* that input — the same odd- n_C quaternionic twist that forces CP supplies it. BST forces what CFS allows.
- Tier + ceiling. Gated on the credential (= the same Φ'). Cal ceiling: continuum-limit / leading-order — never “forces Einstein.” Held behind the outreach vet.

II.3 — CP violation: FORCED (K1303 / T2547) — existence only

- Misalignment leg. The up-type Yukawas are a saturation-inverted tower (top pinned at the Shilov boundary $y_t = 1$; T2514/K764–766); the down-type are a feed-down tower at integer degrees $\{1, 3, 5\}$ (K1012). Different mechanisms $\rightarrow$ not an up $\leftrightarrow$ down mirror $\rightarrow$ generic misalignment $\rightarrow$ the CP-killing cancellation is off the table (F874 Gate A).
- Complex-twist leg (source-verified). Odd $n_C \rightarrow$ quaternionic $Sp(2)$ spinor $\rightarrow$ non-removable $\sqrt{-1}$ (F874 Gate B).
- Two blind confirmations. (a) Operative-spinor pin (F875): pre-registered blind, then computed — the operative eigenvalue of the CP twist is $\omega = e^{\{2\pi i/3\}}$, not -1 . (b) J existence-score (K1304): the construction-independent $\det[H_u, H_d] \neq 0$ / rephasing-invariant $J \neq 0$ is IN — existence banked.
- Tier: Structural, forced — EXISTENCE ONLY. The magnitude is OFF the table: every value of δ_{CP} (or J) is a reverse-fit (K684). ★ And as of 2026-08-22 this is not merely a caution but a retirement: the corpus’s one banked Jarlskog value (T2259) has been struck, and the Guide’s $J = \sqrt{2/50000}$ row is retired as chance-saturated — see Part III.

II.4 — The tiered scorecard (the honest ledger, v1.0)

Tiers are D(erived) / P(artially) D(erived) / I(dentified) / S(tructural) / C(andidate). “Proved” is not a tier in this vocabulary and is not used here.

result	tier
D_IV ⁵ forced modulo one datum (generation count)	honest ceiling — Leg A Derived (Wolf, source-verified); Leg B(i) Derived (T1829); Leg B(ii) Structural
QM from D_IV ⁵ (10/10 Dirac–von Neumann axioms, zero posits)	Derived — input floor shorter than GR’s (T2543). Casey GO.
(3,1) Lorentzian signature from color-triplet irreducibility (T2545)	Derived
CP violation exists (odd-n_C quaternionic spinor, T2547)	Structural, forced — magnitude OFF
dark-energy sign (sub-Poisson, no crossing; T2546)	Structural, forced — magnitude open
$\sin^2\theta_W = 3/13$ (parity-forced shape)	Partially Derived — normalization owed (#79)
★ THE SM HYPERCHARGES (T2573, the mechanism theorem)	DERIVED-CONDITIONAL on I-1...I-4, stated on its face: “Given the OBSERVED CHARGED SPECTRUM (observational input, same class as Five-Absence), the surviving gauge group (T2520), charged Yukawas as the required couplings, and the topology triple, the SM hypercharges are DERIVED — with NO anomaly condition used anywhere — with the ruler as the only free scale; the descent freedom is EXACTLY ONE BIT and it is the ν^c slot — whose removal is the Five-Absence, NOW WITH A MECHANISM.” I-1: G_SM exactly (T2520 + ungauged B–L) · I-2: charged Yukawas only (Majorana bank) · I-3: the observed charged spectrum, OBSERVATIONAL · I-4: the topology triple. Companion: the program’s first EVIDENTIAL anomaly check (toy 5483 — all four sums vanish AND the perturbation control fires: $Y^3 = 91/4500$ at $Y(Q)+1/30$). The gap-b exotic dies by color counting, not anomalies.
$\alpha = 1/N_{\max} = 1/137$ (interior charge/spectral-cap count; not Wyler)	Identified — and the EXPRESSIBILITY argument for it is WITHDRAWN. A pre-registered null with $N_{\max}$ excluded from the pool puts 137 in the bulk (rank 30/76; 23 BST-integer targets score higher), reproduced independently at 27/76. “ <i>The BST integers can express 137</i> ” is what chance supplies at that magnitude, not evidence. The forward form is a mnemonic for a mechanism still owed (III.6a; K1813). The tier is unchanged pending Keeper/Casey — what is struck is one argument for it, not the row.
$m_p / m_e = 6\pi^5$ and the region-matched mass computations	region-matched Derived — dev% is cross-region confirmation, $\frac{7}{8}$ tier (K1245); target-innocence guard applied (K1246)
Koide $Q = \text{rank}/N_c = 2/3$	CONDITIONAL-FORCED — see FLAG H2. The <i>relation</i> stands; every derivation route is now CLOSED (rank: K1221/K1624; raw- ν + exponent: toy 5407; overlap-norm / Lane B: toy 5408 + F506, ruled K1749, amended R62-A 2026-08-23). It is not “Tier-1 exact,” and there is no live gate.

result	tier
PMNS $\theta_{12} = 3/10$	Identified — 3/10 is the geometric-primary FORM (the kernel output, cleaner than the nearer 4/13 or 42/137 alternates); the VALUE is Identified. ¶ This row has NOT been through the Part III discriminating-power test — that the kernel <i>outputs</i> 3/10 rather than the nearer fits is the strongest thing said for it, and it is an argument about provenance, not about N.
gravity emergent / no graviton / holographic four-framework convergence	Framework Evidence at the exhibited-map strength

II.4a — ★ The flavor sector, at its closing tier (K1808 / K1810; ledger **grace_CKM_PARTIALLY_DERIVED_explicit_split_ledger_2008-22.md**) The CKM sector closed on 2026-08-22 at PARTIALLY DERIVED, explicit-split form. It is stated here in that form and no other. Every row of the previous version's CKM block is superseded.

CKM item	status
θ_{12} (Cabibbo) — $V_{us} = 1/\sqrt{20} = 0.223607$	DERIVED (T2530, from the Derived $m_s/m_d = 20$, T2529) — blind, forward, frame-independent. Now pinned to the primary (PDG 2024 Rev 12, cited to its equation): $\lambda = 0.22501 \pm 0.00068 \Rightarrow 1/\sqrt{20}$ sits at -2.06σ , and a <i>non-derived</i> BST form ($2/\sqrt{79}$) sits at $+0.01\sigma$ — a figure that must NOT be quoted bare, because it is reporting a STEP SIZE, not physics. ($2/\sqrt{80} = 1/\sqrt{20}$ exactly, so the two “rival” forms are one expression with and without a -1 on the integer. At $N = 80$ one unit of that integer moves λ by 2.06σ , and the gap it closes is 2.06σ : $\text{step/gap} = 0.996$. A ± 1 on an integer whose natural step matches the discrepancy is a knob with exactly enough range to reach the target — which is what the adjacency penalty was measuring all along. The honest statement is “one bit, two observables, both improve,” and the A leg is the one doing real work: a 4.55σ step closing a 5.07σ gap, landing off bullseye at -0.52σ .) Both facts are reported, and neither moves the tier: fit quality is cross-region confirmation, orthogonal to tier (K1245), and a forward derivation is not demoted by a better-fitting fit. $\Rightarrow$ THE HONEST FORM OF THIS ROW IS A PREDICTION ON ONE SIDE OF A LIVE ANOMALY: with $ V_{ud} = \sqrt{(19/20)}$, $1/\sqrt{20}$ makes the first row EXACTLY unitary by construction, while the direct determinations sum to 0.9984 ± 0.0007 — a 2.3σ shortfall. We predict the unitary value; the measurements currently do not reach it. That can fail, and it is the form this row should ship in. (<i>Framing is Casey's call. Earlier circulated figures of -0.87σ and -6.03σ came from unpinned observables and are withdrawn; the real stake is $\sim 2\sigma$.</i>)
The skeleton — which entries vanish at first order (the cross-parity selection rule)	DERIVED (K1324, three blind routes)

CKM item	status
Flavor-universality = the partial-isometry condition => CKM = $U_{up}^\dagger U_{down}$ is the forced survivor ★ The ORDER of the split — the (1,3) corner is suppressed by exactly ONE power of t relative to the (2,3) subdiagonal, because $0 \rightarrow 4$ takes four rungs on P_6 and $2 \rightarrow 4$ takes two	DERIVED (R52, gated) DERIVED — every candidate series collapses to $G _{\text{even}} = \beta S + \alpha S^2 + \gamma \cdot 1$ (char. poly $\lambda^3 - 5\lambda^2 + 6\lambda - 1$; Cayley–Hamilton residual 0.00e+00), corner ratio $t/(1+4t)$, γ genuinely absent. => “ <i>why is V_{ub} so much smaller than V_{cb}?</i> ” is answered, not fitted — no fitted integer appears anywhere in it
CP violation — existence V_{cb}	DERIVED (T2547) — existence only VALUE RETIRED, POSITION KEPT. The value’s license was “a ~5% match against ~5%-uncertain data” and the data sharpened: 8.9σ vs exclusive, 3.61σ vs the independently-pulled inclusive $42.16 \pm 0.51 \times 10^{-3}$, outside the union band at both ends, with no pre-registered side. Kept at Identified: V_{cb} is a down-sector-only reading because the up 23-mode refracts past the boundary (radius $1.225 > 1$) and vanishes $\rightarrow$ the top decouples. <i>A position has no measurement as its license, so sharper data cannot kill it.</i>
The magnitude of the split	OPEN. Observed ratio $\approx 0.093 \Leftrightarrow t \in [0.120, 0.190]$ at P_6 . The ORDER is derived; the VALUE is not.
δ_{CP} magnitude > COUNT: 1 of 4 CKM parameters DERIVED. 3 OPEN.	OPEN (T2547 banks existence only)

★ The sealed negative — a banked asset, not an absence. Five candidate series for the split magnitude were sealed (SHA256 43ad5eb3...f43488, verified intact) before the target band was scored, and opened only after the series were filed in writing. All five MISS the band $[0.081, 0.108]$, all high, none close: 0.2500 (2.65 $\times$), 0.2000 (2.12 $\times$), 0.3276 (3.47 $\times$), 0.3158 (3.34 $\times$), 0.3571 (3.78 $\times$). This is the strongest form of negative the program produces — the procedure was frozen before the number existed. (*The procedure-freeze rule itself came out of this sector: a bar handed alongside an unfrozen procedure is a tuning channel, not a pre-registration.*)

Convention caveat that must travel with the tier: t is convention-carrying — it rescales as $t \cdot c^2$ under $Q \rightarrow cQ$ — and must not be quoted bare. Quote the invariant $\sigma_\chi(G)$.

Dead lanes, banked as exclusions (do not re-open): radial overlap (proved ceiling, ~1.1M configurations, zero passes) · the one-insertion weak current · the FK ladder (wrong space) · the QM commit operator (wrong space, and a projector) · the Q^5 parity fold (forced and genuine, but a projector) · $P = 1 + \varepsilon Q$ (Q parity-odd => the identity on generation space for every ε) · all five sealed series.

II.5 — Dark energy: the SIGN is forced (Structural), and it is more falsifiable than Λ CDM

Pauli exclusion $\rightarrow$ sub-Poisson statistics $\rightarrow$ the sign of the vacuum fluctuation is forced (Bernstein/complete-monotonicity, T2546). Survives the DESI DR2 radial BAO; shares Λ CDM’s $\sim 3\sigma$ on BAO+CMB. Differentiator (K1298): BST predicts a monotone, one-sign, power-law residual with no crossing — vs Λ CDM’s zero and CPL’s sign-flip crossing. DR3-testable. Λ stays Structural; we never say “BST derives dark energy.”

II.6 — Falsifiers (the theory bets)

The Five Absences — no GUT, no proton decay, no dark-matter particle, no magnetic monopole, no sterile neutrino, no SUSY spectrum. Plus the DUNE / neutrino-ordering predictions and the dark-energy differentiator. A positive detection of any Absence refutes the framework.

Honesty note on the word “falsifiable” (standing rule, applied here): a prediction with a direction but no size cannot fail. Of the falsifiers above, the Five Absences and the DE no-crossing signature carry a decidable yes/no; the neutrino-ordering row does too. We do not count as falsifiers any result that is covariant-by-construction or that names a direction without a magnitude. Report the can-fail count, not the denominator.

Part III — THE DISCRIMINATING-POWER STANDARD (new in v1.0; absorbs K1809)

This is the part of the paper a hostile referee reaches first, so it is stated first-person and without cushioning.

III.1 — The problem, stated against ourselves

BST’s algebra is closed over five small integers (Part 0.5). A closed algebra over $\{3, 5, 6, 7, 137\}$ and rank 2 generates many simple expressions inside any given error bar. So the observation “*quantity X equals a simple BST integer ratio, to 0.4%*” is, on its own, worth nothing until one further question is answered:

How many other equally simple BST forms also sit inside that error bar?

If the answer is “several,” the match has no discriminating power: the theory did not predict the number, it merely failed to exclude it — along with its neighbours. The number of competitors is part of the result, and a result reported without it is incomplete.

III.2 — The test, and the honest reporting form

For any BST-integer form banked against a measured band: 1. Generate the competitor pool at the published form’s own complexity class (not below it). 2. Count the members that land inside the band. Call it N. 3. Score N against the null — how many forms of that class does chance put in a band that width? Then report “smallest-of-N indistinguishable BST forms, N reported” — never a bare match. This form is available only where the family is enumerable and theory-generated; where the count is chance-saturated the only honest disposition is retirement (III.4a). 4. If a competitor of equal simplicity fits better than the published form, that fact is reported in the same sentence.

And the instrument itself must be positive-controlled. A pool that generates only simple forms (a , $\sqrt{a}$, a/b , $\sqrt{a/b}$, $a \cdot b$, $1/(2\sqrt{a})$) cannot clear a compound form, and its silence on one is a limitation of the instrument, not a clearance. A search that cannot succeed proves nothing.

III.3 — We ran it on ourselves first, and it retired two theorems

T2198 (CKM magnitudes as $1/N_{\text{max}}^2$ integer ratios) and T2259 (the Jarlskog and Wolfenstein $\bar{\eta}$ rows) were retired by their own author on three grounds:

1. No discriminating power. The band admits four consecutive integers in the numerator, three of them “BST integers,” all indistinguishable.
2. The fit drifted with the data. T2198 quoted obs $|V_{cb}| = 0.0411$ for its 0.4% agreement. Against current PDG 0.0418 the same numerator (770) is 1.9% off, and the best member of its own integer family is now a different numerator — $784 = \text{rank}^4 \cdot g^2$, at 0.117%. A relation whose winning integer changes when the data moves is a fit.
3. The target sharpened past it — 8.9σ against the exclusive determination, 3.61σ against the independently-pulled inclusive one ($42.16 \pm 0.51 \times 10^{-3}$), outside the union band at both ends, with no pre-registered side.

(Grounds 2 and 3 are statements about the target moving and about distance from named determinations that carry error bars — not fit-quality comparisons between two forms at one unpinning target. That distinction is the subject of the box in III.4, and it is why these two survive the withdrawal of the $\bar{\eta}$ ranking line.)

(Grace R53; ratified Keeper K1795. Registry rows and graph nodes carry retirement markers as of 2026-08-23.)

III.4 — Then we swept the siblings, and the sweep needed a null model before it could rule

A retirement is a loaded string: the standard that retires one result must be run across everything that shares it. Applied to the five remaining CKM/CP formulas in the curated Guide (K1809), then re-scored against a null (K1809-B, Cal §705) — because a competitor count is not a verdict until you know how many competitors chance supplies.

quantity	published BST form	value	competitors in band	chance expectation	verdict
γ (CP phase)	$\arctan(\sqrt{n_C})$	65.91°	10	—	RETIRE
$\bar{\eta}$	$1/(2\sqrt{2})$	0.35355	4	—	RETIRE
J_CKM	$\sqrt{2/50000}$	$2.83e-5$	41	35.7	RETIRE — saturated
$\bar{\rho}$	$1/(2\sqrt{(2n_C)})$	0.15811	2	~ 3	NON-DISCRIMINATING (not failed)

γ admits ten indistinguishable readings inside one error bar.

★ Why this table carries no σ column, and no “which form fits best”

A competitor COUNT is target-independent. A competitor RANKING is not. An earlier draft of this section led with its sharpest-looking line — *for $\bar{\eta}$ a competing BST form of equal simplicity fits about five times better than the one we publish*. That finding REVERSES under a different determination of the same observable: against one target the competitor wins and the published form is the worse member; against the other, the published form wins and the competitor is the loser. The comparison was inheriting every uncertainty in a target that was never pinned — and worse, our own data layer carried two mutually inconsistent targets for that one observable, so each form could be paired with the target that flattered it. (*Root cause found and flagged; see below. Cal caught the reversal; Keeper wrote the finding and withdrew it; the ranking line is HELD until the target is pinned to a named determination with a stated error bar.*)

The retirements do not need it. In-band counts are near-constant across band centres in this decade, so saturation is target-independent by construction — the counts rule, the σ ’s were decoration, and the decoration was the part that was wrong. Drop the σ , keep the retirement.

★ RESOLVED, at the primary — and the resolution went against BOTH earlier readings. Pinned to PDG 2024 Rev 12 with each determination named: $\bar{\rho}$ — the published form is best (-0.10σ against 0.1591 ± 0.0094); $\bar{\eta}$ — the published form is best ($+0.17\sigma$ against 0.3523). So the original “*a competitor fits $\sim 5\times$ better*” is refuted, and the later reversal of it is also refuted: the answer was neither, and both readings had been scored against unpinned targets. Five rankings in circulation that day changed. ZERO counts changed, and every retirement stands — which is the whole argument for preferring count instruments. (*Cal, pass 2, primary-pinned.*)

Two distinctions the null model forces, and both cut against us: - $\bar{\rho}$ is not a failure — it is empty. Two competitors where chance supplies about three is not a family the theory narrowed; it is a band too wide to distinguish anything. Two is not one, and it is still not evidence. Reporting $\bar{\rho}$ as “failed” would over-state the finding in the same way the original row over-stated the match. - J is TESTED, and it does not survive. Its band holds 41 in-band forms against a chance expectation of 35.7 — saturated. The theory’s form is one of ~ 40 things that fit, which is what noise produces on its own.

=> γ , $\bar{\eta}$ and J are RETIRED, and the retirement takes a specific form this paper had to be corrected on — see III.4a. That reconciliation is a dispatch gate and is tracked separately (K1809-B / K1801, Cal + Casey GO).

III.4a — ★ Why “smallest-of-N, with N reported” is NOT available here

The honest reporting form of III.2 — “*smallest of N indistinguishable BST forms*” — presupposes two things: that the family is enumerable, and that choosing within it is a meaningful act. Both hold when a theory narrows a set of candidates and picks one. Neither holds when N is simply what chance supplies.

Naming N when N is the noise level dresses noise as bounded ambiguity. It reads as “*we have narrowed this to one of four,*” when the truth is “*the band is wide enough that four things fall in it, and ours is one of them.*”

This paper carried the wrong option and was corrected on it. The rule, restated: *smallest-of-N* is available only where the family is enumerable and theory-generated. Where the count is chance-saturated, the only honest disposition is retirement. The distinction is decided by the null — which is why the null is not optional.

III.5 — Calibration against ourselves: the first J verdict was an instrument failure, in BOTH directions

J is this section’s own worked example of a competitor count going wrong twice.

First reading — “zero competitors.” The pool generated simple forms only (a , $\sqrt{a}$, a/b , $\sqrt{(a/b)}$, $a \cdot b$, $1/(2\sqrt{a})$); J’s published form is compound ($50000 = n \cdot C^5 \cdot (2^{\wedge} \text{rank})^2$), a complexity class the pool never reached. Its silence was a limitation of the instrument, not a clearance — and reporting it as a pass would have been exactly the false-negative error this section exists to catch. So the first verdict recorded was “UNTESTED.”

Second reading — untested was also wrong. A pool built to J’s own class returns 41 in-band forms. But the count alone still rules nothing: against a chance expectation of 35.7, J’s band is saturated. The correct verdict is neither “passed” nor “untested” — it is TESTED AND RETIRED.

Both errors were the same error at different signs: reading an instrument’s output without knowing what that instrument does on nothing. A pool that cannot reach the form under-counts; a pool with no null over-interprets whatever it does count. A competitor count is not a verdict until it is scored against chance — and a null with a free sampling range is not a null until that range is swept. Cross-check with an estimator that shares none of the first one’s free choices.

III.5a — The same standard from the other side: a gate that cannot FAIL (Cal §702–§704, Lyra, Elie; 2026-08-23)

Part III asks “*could a competing form have matched?*” The mirror question is “*could this gate have failed?*” — and it caught a live BST result the same week, in the Koide lane.

The proposed gate was the scalar equality $R = (\Sigma s)^2 / \Sigma s^2 = 3/2$ on three amplitudes read off the Bergman norms. Write $s_{\nu} = c \cdot \|f_{\nu}\|^p$. R is scale-free in c — but not in p:

$R \rightarrow 3$ as $p \rightarrow 0$, $R \rightarrow 1$ as $p \rightarrow \infty$, and R is strictly monotone in p. So for ANY three distinct positive norms a root $R = 3/2$ exists — and it is UNIQUE. (*Monotonicity verified: 0 violations in 20 000 triples, independently reproduced at worst increment 0.00e+00. Demonstration triples, drawn at random and labelled as such: $p^* = 4.44223, 6.95378, 1.80930$ — each landing 3/2 to 12 digits.*)

The gate’s can-fail count was ZERO. With the exponent unpinned it returns the target for any input, so passing it would have proved nothing about D_{IV}^5 . And uniqueness sharpens the diagnosis rather than softening it: because the root always exists and there is exactly one of it, a gate of this shape is never “*does the number come out*” — it always does, at exactly one p. It is only ever: does the PINNED p equal p^* ?

★ And p^* is not a number of the theory — it spreads $\sim 100\times$ across candidate norm objects. So the look-elsewhere here is TWO-DIMENSIONAL: {which norm OBJECT} $\times$ {which EXPONENT}. Pin both, in writing, before computing, or a two-draw result is being reported as one.

What follows, in the order the evidence actually runs:

1. The negative stands on its MECHANISM, and needs no counterfactual. $R = 3/2$ demands a max/min amplitude ratio of 22.96. Every form finite at the forced ρ -addresses delivers less: 8 of 8 finite evaluations miss HIGH, range [1.793, 2.779], zero low. A one-sided miss is a systematic, not scatter — the overlap norm at

the lepton addresses is uniformly under-hierarchical. That is the result, and it is stated without appeal to any capability argument.

2. Then the attribution, with its basis stated. Capability of the family that produced the negative: can-fail = 2 of 12, the survivor (F5) clearing 1.5 by 0.0159 on 0.0037% of its domain. And the caveat travels with it: that capability was measured by sweeping ν over a domain the theory forces to exactly three fixed values — the best available proxy, used as a proxy and labelled as one. (*The tested family is five distinct forms, not six: F2 $\equiv$ F4 algebraically, verified to 45 decimal places — two can-fail channels, not three.*)
3. The ordering above is itself a correction this paper needed. An earlier draft validated the negative using the floating-p family's capability — an instrument this very section had just proved cannot fail, and not the instrument that produced the negative (which ran at pinned p). A control that PASSES can be the wrong control: the instrument validated must be the instrument used.

Two rules survive, and the second is the one worth keeping: - Count free parameters against targets before scoring a match. One free coordinate and one target is a reparameterization, not a prediction. (*A form that fits any target is definitionally empty — distinct from a form merely awaiting confirmation.*) - ★ Over-stating a NEGATIVE is the same error class as over-stating a derivation. A non-detection is worth nothing unless the instrument that produced it was shown capable — but the capability demonstration must be of that instrument, and a mechanism that explains the miss outranks any capability count. Lead with the mechanism; attribute second; state the proxy.

The residual obligation is a provenance question, and it is the right one: pinning the exponent to the quark toy (T2529) is an *inherited* constraint only if that pin was target-innocent — *when the quark toy was fixed, how many members of the map family were tried?* One, written down first $\Rightarrow$ the pin is clean. Several, keeping the one that gave $m_s/m_d = 20 \Rightarrow$ any later gate of this shape is a second draw on one free coordinate. (*The one-sided 8/8-high miss is corroboration that the pin was not tuned to the lepton sector — an input to that audit, not its answer.*)

III.5b — ★ THE INSTRUMENT'S OWN BLIND SPOT, and the quantitative floor it produces

A pilot across 39 banked rows gives the test a scale, and the pool-independent statistic is the RATIO count/chance, never the absolute count — the same structural move as *a count is target-independent*:

claimed agreement	in-band count / chance	reading
$\leq 0.006\%$	band essentially EMPTY	evidence
$\geq 0.1\%$	1.00 ± 0.2 , UNIFORMLY — across four pools spanning a 425× range	SATURATED

The crossover sits near 0.01–0.05%. (*Provisional: a second, independent instrument is pending, and the threshold number is not banked until it lands.*)

★ AND THE CLAUSE THAT MUST TRAVEL IN THE SAME SENTENCE — a discriminating-power count CANNOT SEE TARGET-INNOCENCE

For a form TUNED to a target, the band is narrow and therefore EMPTY BY CONSTRUCTION. The instrument measures whether a band is crowded. It never measures whether the form earned its position in that band. A fitted six-digit form and a forced six-digit form return the identical in-band count of zero.

So the honest statement of the standard is one sentence with its condition attached, not two:

BST treats agreements tighter than $\sim 0.01\%$ as evidence, and agreements near 1% as consistency checks — PROVIDED THE FORM WAS FIXED BEFORE THE TARGET WAS CONSULTED.

That proviso is not a bolt-on; it is III.6's distinction restated in the instrument's own units. (*SUPERSESSON NOTE, 2026-08-24: the anomaly-freedom check ("W2"), excluded as construction-guaranteed when the hypercharges were anomaly-SELECTED, is now EVIDENTIAL: T2573's custodial route derives the charges with no anomaly input,*

and toy 5483 verifies all four anomaly sums vanish while the perturbation control fires loudly. The exclusion was correct when made and is correctly lifted — both facts on the record, in that order.) The test and the provenance requirement are two halves of one criterion, and either alone is defeatable: provenance without the count admits crowded bands, and the count without provenance admits tuned ones.

★★ THE ASYMMETRY THAT FOLLOWS, and it is why III.6a still stands

This instrument's FAILURES are robust; its PASSES are conditional. A crowded band is crowded whatever the form's provenance — no innocence argument rescues α , because 23 targets scoring higher is a fact about the neighbourhood, not about how we arrived. But an empty band is only evidence if the form got there without looking. $\Rightarrow \alpha$'s failure is final on this instrument alone. The muon's pass is not — it is a pass conditional on target-innocence, and it removes ONE objection from the four-part K968 bar {exact-argument, ADDRESS, exponent, sector}. The muon stays IDENTIFIED and promotes nothing.

III.6 — What the test does NOT touch, and why that is the argument

Nothing derived survives by permission of this test — the derived results are immune to it by construction, and that is the point: - $V_{us} = 1/\sqrt{20}$ comes forward from the Derived mass ratio $m_s/m_d = 20$ (T2529). It was computed blind, before comparison. There is no band-search anywhere in it. - The ORDER of the $|V_{ub}|/|V_{cb}|$ split uses no fitted integer at all — it counts rungs on P_6 . A competitor pool has nothing to compete with. - The parity forcings (odd $N_c \rightarrow$ the $\sin^2\theta_W$ shape; odd $n_C \rightarrow$ CP exists) are qualitative. There is no second BST expression that also makes a representation quaternionic. - The sealed negative was frozen before the target existed. Look-elsewhere cannot reach it.

III.6a — ★ THE TEST RUN ON OUR TWO MOST FAMOUS NUMBERS: one fails, one survives

The claim above — *that this test prunes coincidences and cannot touch forward computations* — is the kind of claim a referee is right to distrust, because a filter built by the people it judges tends to pass what they like. So it was run, on the same day with the same instrument, on the two most prominent numbers in the corpus. Neither was chosen to make this section look good, and the famous one failed.

	$\alpha^{-1} = N_{\max} = 137$	$m_\mu/m_e = (24/\pi^2)^6$
instrument	pre-registered null over BST integers, $N_{\max}$ excluded from the pool	same null, same day, 4000 targets
result	137 sits in the BULK — rank 30/76 with 23 targets scoring higher; independently reproduced on a different counting convention at 27/76 with 26 higher, finding the same two attractors ($128 = 2^g$, $125 = n_C^3$)	0.003441%; four decoy targets get ZERO hits, the muon gets ONE and it is the banked form; $p = 1.52\%$ on one pool, 0.75% on the other
verdict	FAILS. “BST integers can express 137” is not evidence — it is what chance supplies at that magnitude	DISCRIMINATING

$\Rightarrow \alpha$'s forward form is a MNEMONIC for a mechanism still owed. The number is not refuted; the expressibility argument for it is. (Elie toys 5457/5458; K1813/K1814.)

Three conditions travel with the muon p-value, and they are binding: 1. The tolerance was set by the form's OWN accuracy, so a tighter form gets a smaller p by construction. p measures the rarity of agreement this good within

this family — not the probability the form is right. 2. “ $24 = N_{cl} \cdot 2^{\text{rank}}$ ” is untested and probably generic. The null integrates over A, k and B, so 24’s genericity is priced in — but the verdict is not robust to any claim that 24 is *forced*, which is a separate and untested claim. 3. The K968 bar is unmoved. It requires {exact-argument, ADDRESS, exponent, sector}; this supplies discriminating power only — one objection removed from four. The muon stays IDENTIFIED.

(RATIFICATION NOTE, 2026-08-24: the split-rank result and its dependents — hedged as “the conditional stack” when filed — are now RATIFIED (Shapovalov instrument, sealed-miss protocol). The hedges in this document are lifted where they pointed at that stack. And the ratification produced a new structural-identity candidate: THE EDGE IDENTITY — w_0 = the spinor edge = $(n-1)/2$, family-swept 4/4 — carried with its literature-pin caveat on its face: the Dirac-bound page pin is pooled for a book-hour and the identity does not bank before it.)

(Reporting standard, from the same round: quote the in-window count, not the pool size — two pools giving $p = 0.75\%$ and 1.52% differ in window occupancy, not quality, and their first-order expectations of 0.62% and 1.34% match. Pool size is not comparable across instruments.)

Why this is the section’s strongest support rather than its most damaging result: - The same instrument returned opposite verdicts on two results of the same house style — so it is not a filter that passes what we like. It discriminates among our own claims. - It failed the one we are most attached to. A test that had cleared α would have told us nothing about the test. - And the survivor survives for the reason III.6 predicts: the muon form is a forward computation whose competitors were enumerated and beaten, exactly as V_{us} is immune because it was computed blind before comparison. The prediction of this section was that coincidences would prune and forward derivations would not. Both halves happened, on the hardest available case, in a test its author did not choose.

★ The contrast IS the paper

Every result this test killed was a numerical coincidence found by looking at a measurement. Every result it cannot touch was computed forward from the geometry before the measurement was consulted. The test does not merely prune the corpus — it separates the two kinds of claim we have been making, and the separation runs exactly along the line between fitting and forcing. A framework that can retire its own flagship-adjacent results under its own rule, and whose *derivations* are structurally immune to that rule, is making a different kind of claim from one that reports matches.

III.7 — Standing obligations created by this section

1. No BST-integer match enters this paper, the Guide, or the registry without its N — and no N is a verdict without its null. How many forms of that class does chance put in a band that width?
2. Where the count is chance-saturated, retire. *Smallest-of-N* is available only for an enumerable, theory-generated family (III.4a).
3. Any new retirement triggers a sweep in both directions — what the retired result was load-bearing for, and what shares its name (killing a string voids its live namesakes).
4. Every competitor pool is positive-controlled on a must-catch and a must-reject case, and must reach the published form’s own complexity class, before its output — silence included — is read as anything. 4a. PIN THE TARGET BEFORE ANY COMPARISON, to a named determination with an error bar. A count survives an unpinned target; a ranking does not. No σ may be quoted from a source that carries no error bar — and where a source carries *two* targets for one observable, every fit-quality statement drawn from it is unfalsifiable by construction, because a form can be paired with the target that flatters it.
5. Every gate is scored for can-fail before it is scored for pass/fail — count free parameters against targets, and count the look-elsewhere dimensions (object *and* exponent). A gate that returns the target for any input is empty.
6. Every negative leads with its mechanism, attributes second, and states the proxy in any capability claim. The instrument validated must be the instrument used. Over-stating a negative is the same error class as over-stating a derivation.

(A seventh candidate — “any predicate you author is a candidate false-negative machine” — is not a separate rule. It belongs to obligation 4 (and to the existing §698 guard) as a scope note: the failures that prompted it were filters written that session and never positive-controlled. The finding was that the rule was not applied, not that another rule was needed. Ruling: Lyra.)

The claim, exactly (v1.0)

Reality is forced to be D_{IV}^5 modulo one empirical datum — the number of generations (one measured number in, the rest forced — GR’s standard of *input economy*, with the honest, named I.5 residuals still open; not a closed unconditional proof). The object reproduces quantum mechanics (10/10 axioms, zero posits) and a named, bounded part of the Standard Model’s dimensionless structure with no fitted dimensionless parameters — *and the price of “no fitted parameters” is the open list in Part R, which is stated as part of the claim, not after it*. It is the independent convergence point of four fundamental frameworks and forces what one of them (CFS) can only allow; and it makes sharp falsifiable predictions (the Five Absences; the dark-energy differentiator). Every numerical match is reported with its discriminating power or not reported at all (Part III) — a standard under which two of our own theorems were retired and three more curated rows failed. Every claim is exhibited, every tier labelled, every residual a named challenge. BST = Tegmark’s MUH + a selection principle.

Part R — Residuals, collected (the honest challenge list, v1.0)

- Leg B(ii): does the color-dimension condition select D_{IV}^5 *uniquely* within the rank-2 census (vs E III)? — group-theoretic uniqueness step open.
- I.1 innocence: the $m_s = 3$ n-scan (independent verify) + R3/R5 criteria-innocence.
- CKM, 3 of 4 OPEN: the split magnitude ($t \in [0.120, 0.190]$, convention-carrying — quote $\sigma_\chi(G)$); the δ_{CP} magnitude; V_{cb} ’s value (retired, position kept).
- CP magnitude: off the table by ruling (K684); the one banked Jarlskog value is retired (Part III).
- Λ magnitude: open (only the sign is forced).
- $\sin^2\theta_W$ normalization: PD $\rightarrow$ Derived promotion-path (#79).
- The soft up-mass m_u : the one genuinely soft mass (no forced form, K1016/F890); the up-charm 12-block $= \sqrt{(m_u/m_c)}$ is run in reverse — a consequence of CKM + down, not a forward input. Forcing it forward would be fitting the soft mass.
- PMNS $\theta_{12} = 3/10$: Identified, and not yet through the Part III test (N unknown).
- The $\bar{\eta}$ RANKING claim: RESOLVED at the primary — the PUBLISHED form is best ($+0.17\sigma$). Both earlier readings were wrong, in opposite directions, and both had been scored against unpinned targets. Held correctly until pinned. (*Formerly: HELD in both directions.*)
- ★ The Cabibbo/ V_{us} observable: UNPINNED, and worth $\sim 5\sigma$ to our one Derived CKM row. See the II.4a scorecard note. Three tier-D forms for one piece of physics, three different targets, no error bars.
- The data layer’s contested targets: FLAGGED — and this does NOT contradict Part III’s primary-pinned resolution. The primary pin (PDG 2024 Rev 12) resolved the ranking question (which form fits best); this flag records the file state (two stored targets per observable, no winner chosen in the data layer) pending the landing pass that writes the pinned determinations into the rows. Resolution exists; landing is a separate, owned step. Five observables carry two mutually inconsistent targets in `data/bst_constants.json` ($\bar{\eta}$ 0.349 vs 0.357 · $\bar{\rho}$ 0.159 vs 0.150 · Jarlskog $3.08e-5$ vs $3.0e-5$ · proton $\mu \cdot \alpha$ binding), and 193 of 197 rows carry no error bar at all. Rows are marked `CONTESTED / sigma_quotable: false` with both targets recorded; no winner has been chosen, because choosing one without a source would be the fit the flag exists to prevent.
- The Guide’s $\gamma / \bar{\eta} / J$ rows: RETIRE (chance-saturated; *smallest-of-N* is not available — III.4a). $\bar{\rho}$: re-state as NON-DISCRIMINATING, not failed. Dispatch-gated on the K1809-B / K1801 reconciliation.
- CFS credential: $= \Phi'$ (gated; Cal ceiling continuum-limit).

Each is concrete and named. The residual list is the paper’s contribution as much as the results — it is the FOREVER-living rigor standard.

🔗 FLAG H2 — the Koide row (Grace, v1.0; Lyra/Keeper to ratify the wording)

v0.2 carried Koide as “Tier-1 exact.” Two problems: “Tier-1” is not a tier in the D/PD/I/S/C vocabulary this paper enforces on everything else, and the forcing is falsified, not open. K1221/K1624 killed the rank route; toy 5407 killed the raw- ν + exponent route; toy 5408 + F506 killed the overlap-norm route, and K1749 closed the mass gate on 2026-08-21 — a closure I initially recorded here as a live lane and corrected the same day (R62-A). The relation stands as a relation; no route derives it. I have re-stated the row as CONDITIONAL-FORCED, the tier K1749 assigned.

The residue lead (the ν -slice mismatch — T2529 is fixed- ν / varying-degree, the leptons are varying- ν / fixed-degree) is flagged, NOT a live gate, and this paper does not carry it as one. If a future pass reopens it, the win condition is “Koide’s *tilt* is Derived” — not the phase δ , not the scale M — counting once with T2003. That is stated here so it cannot be widened later.

Version history

v0.2 → v1.0 (Grace, 2026-08-23). (1) K1809 absorbed as Part III — the discriminating-power standard, the self-retirement of T2198/T2259, the sibling sweep (γ 10 / $\bar{\rho}$ 2 / $\bar{\eta}$ 4, with $\bar{\eta}$ ’s better-fitting competitor stated), the J-unttested calibration, and the four standing obligations. (2) The whole CKM block replaced by the sector’s closing state (K1808/K1810): 1 of 4 Derived, the ORDER result added, V_cb value-retired/position-kept, the sealed negative banked as an asset, the dead lanes listed. (3) Part 0 headline scoped (FLAG H1) — “no fitted parameters” stated together with what it cost. (4) Koide re-tiered (FLAG H2). (5) PMNS θ_{12} marked as not yet through the Part III test. (6) Falsifier honesty note added — report the can-fail count. (7) Convergence counted per independent fact, not per observation. (8) “Proved” removed as a tier word. (9) Part 0.5 now names the vulnerability its own closure creates, which Part III answers.

v1.0 amendment (same day, R62-A): Koide re-stated from “live Lane B gate” to CONDITIONAL-FORCED, all routes closed (K1749); Part III.5a added — Cal §702’s can-fail = 0 result as the mirror of the discriminating-power test, with the negative-validation corollary; standing obligations 5–7 added. (*The correction is itself an instance of obligation 7: the “never run” claim it replaces was a number-range check read as coverage.*)

v1.0 → v1.0.1 (Grace, 2026-08-23, R63 — seven folds from K1811 + K1809-B/Cal §705, plus Lyra’s ruling on obligation 7). (1) III.5a re-ordered to lead with the MECHANISM — the 22.96 amplitude-ratio demand and the one-sided 8/8-high miss carry the negative; no counterfactual needed. (2) Attribution second, with its basis stated — can-fail 2 of 12, F5 clearing by 0.0159 on 0.0037% of its domain, and the ν -sweep named as the proxy it is. (3) The four exhibited roots deleted (invented triples, none a Bergman norm); three seeded random triples substituted and labelled random. (4) “Six forms” corrected to FIVE (F2 $\equiv$ F4 to 45 dps). (5) $\bar{\rho}$ moved off the fail list to NON-DISCRIMINATING (2 in band, chance $\sim$ 3). (6) J re-scored TESTED and RETIRED (41 in band vs chance 35.7, saturated); III.5 rewritten as a two-sign instrument failure. (7) “Smallest-of-N” withdrawn for $\gamma/\bar{\eta}/J$ — new III.4a: naming N when N is the noise level dresses noise as bounded ambiguity; chance-saturated $\Rightarrow$ retire. Obligations re-cut to six, with the seventh folded into obligation 4 as a scope note. III.6 untouched.

v1.0.1 → v1.0.2 (Grace, 2026-08-23, K1809-C). The $\bar{\eta}$ “fits $\sim$ 5 $\times$ better” line is WITHDRAWN and the σ column is gone from III.4. The ranking reverses under a different determination of the same observable; the counts do not move. Added the governing distinction — *a competitor count is target-independent, a competitor ranking is not* — plus the density control that makes the retirements target-independent by construction, obligation 4a (pin the target to a named determination; never quote σ from a source with no error bar), and the data-layer root cause. The three retirements stand on their counts, unchanged.

v1.0.2 → v1.0.3 (Grace, 2026-08-23, R64). (1) The V_us Derived row now carries the observable-identity flag — λ vs $|V_{us}|$ are different observables, the gap is the Cabibbo anomaly, and the same form runs from -0.87σ to -6.03σ across the readings. The derivation is untouched; what is unpinned is what it is a derivation of. The row

does not ship until one observable and one named determination are pinned. Digits explicitly unbanked (web-search provenance). (2) The $\bar{\eta}$ ranking is held in BOTH directions — neither data-layer row carries an error bar and the one that favours us carries no source, so “the published form wins” is as unsupported as “the competitor wins.”

v1.0.3 → v1.0.4 (Grace, 2026-08-23). (1) ★ New III.6a — the test run on α and the muon, same instrument, same day, opposite verdicts: α ’s expressibility argument fails a pre-registered null with N_{max} excluded (137 in the bulk, rank 30/76, reproduced at 27/76), the muon form is discriminating ($p = 1.52\% / 0.75\%$, four decoys at zero hits). The section’s own prediction, demonstrated on the hardest available case in a test its author did not choose. (2) The α scorecard row carries the withdrawal — the expressibility argument is struck, the tier is not mine to move. (3) The $\bar{\eta}/\bar{\rho}$ ranking question RESOLVED at the primary (PDG 2024 Rev 12): published forms best in both, both earlier readings refuted, five rankings changed and zero counts changed.

v1.0.4 → v1.0.5 (Grace, 2026-08-23, R65). (1) The V_{us} Derived row is pinned to the primary — $\lambda = 0.22501 \pm 0.00068$, $1/\sqrt{20}$ at -2.06σ , a non-derived BST form at $+0.01\sigma$; both reported, neither moves the tier, and the row is re-framed in its falsifiable form: exact first-row unitarity by construction against a 2.3σ measured shortfall. Earlier $-0.87\sigma/-6.03\sigma$ figures withdrawn as unpinned. (2) New I.5a — the commitment route audited to its floor (K1815): a two-step derivation, one irreducible posit (rank = 2, reducible only to minimality — a selection principle), and a cited-not-banked classification step. The chain reaches D_{IV}^n , never D_{IV}^5 . (3) Three guards recorded: one fact one vote (rank = 2 is already a banked integer — never two legs); three namings are one argument; isotropy does no rank work.

v1.0.5 → v1.0.6 (Grace, 2026-08-23, K1817 / Cal §712). (1) New III.5b — the instrument’s quantitative floor AND its blind spot, in one place because they are one criterion: $\leq 0.006\%$ empty $\Rightarrow$ evidence, $\geq 0.1\%$ saturated at ratio 1.00 ± 0.2 across four pools spanning $425\times$, crossover $\sim 0.01\text{--}0.05\%$ (threshold NOT banked — second instrument pending); and a discriminating-power count cannot see target-innocence, because a tuned form’s band is empty by construction. The standard is stated as one sentence with its proviso attached. (2) ★ The asymmetry recorded: failures are robust, passes are conditional — α ’s failure needs no innocence argument, the muon’s pass does. (3) The muon’s three binding conditions added to III.6a, including that it removes one objection from the four-part K968 bar and promotes nothing; in-window count adopted as the reporting statistic over pool size.

v1.0.6 → v1.0.7 (Grace, 2026-08-23, R67). Correction to my own scorecard: the “ $+0.01\sigma$ ” I recorded for $2/\sqrt{79}$ is a step-size artifact, not a better fit. $2/\sqrt{80} = 1/\sqrt{20}$ exactly, so the two forms are one expression with and without a -1 ; at $N = 80$ the integer’s own step is 2.06σ against a 2.06σ gap (step/gap = 0.996). Quoting it bare would have made a knob look like a result. The row now states it as “one bit, two observables,” names the A leg as the informative one, and keeps the derived row’s tier unchanged.

v1.0.7 → v1.1 (Grace, 2026-08-24 — the R93 fold; Cal’s five-item list executed verbatim). (1) T2573 enters the II.4 scorecard in its exact conditional form, statement verbatim, I-1...I-4 and the evidential-anomaly companion on its face. (2) W2’s supersession note in III.5b: the exclusion was correct when made and is correctly lifted — both on the record, in order. (3) I.5a superseded-floor banner: binarity is a THEOREM ($P1' + \text{Lemma } 2$), P2b struck; yesterday’s posit-table kept for the chain; the dimension separation stands. (4) The Sunday stack un-hedged (ratified, sealed-miss protocol) and the edge identity ($w_0 = (n-1)/2$, 4/4 family-swept) entered as a structural-identity candidate with its literature-pin caveat on its face. (5) The 228/360 tension defused in one sentence: the primary pin resolved the RANKING; the data-layer flag records the FILE STATE pending landing — resolution exists, landing is separate.

v1.0 is DRAFT COMPLETE — no open markers. Gates before any external step: Cal cold-read (hostile) + Keeper consistency PASS + Casey GO. NOT external. Nothing pushed.